Aryan Naik

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EDUCATION

MEng Robotics and Artificial Intelligence | University College London (UCL) | 2025 - 2029

Relevant coursework: computational modelling, dynamics and control, embedded systems, electronics, engineering mathematics, thermodynamics and heat transfer, aerodynamics fundamentals

PROJECTS

Aerospace Avionics Firmware - Student Rocketry Telemetry Payload

- Delivered autonomous parachute deployment and descent control across all simulated flight phases by implementing a finite state machine in C++ on a dual-MCU embedded architecture, achieving zero missed state transitions across full end-to-end ground testing
- Enabled reliable real-time inertial data acquisition for attitude estimation and altitude logging by integrating an IMU and barometric sensor over I2C with interrupt-driven execution, sustaining consistent sensor polling at the required sampling rate throughout all test runs
- Validated hardware-software co-design for a flight-ready avionics payload by implementing UART and SPI serial communication protocols with power budgeting and signal conditioning, resolving all identified signal integrity faults prior to integration sign-off
- Applied defence electronics and guidance, navigation and control principles within a safety-critical rocketry context by designing autonomous recovery logic, producing a flight computer architecture suitable for launch vehicle deployment

Full-System Robotics Hardware Integration - Competition Platform

- Produced a fully functional robotic platform that met all competition acceptance criteria by integrating microcontrollers, motor drivers, actuators, and power electronics from breadboard schematic through to physical assembly and closed-loop firmware control
- Achieved verified signal integrity across all subsystems by performing systematic fault-finding and hardware debugging at each integration stage, clearing every functional test checkpoint before final competition submission
- Reduced mechanical iteration time during rapid prototyping by designing structural components, enclosures, and mounting geometry in Autodesk Fusion 360 with explicit tolerancing, enabling direct handoff to 3D printing without rework on final assemblies

SKILLS

Embedded and Firmware: C++, dual-MCU architecture, interrupt-driven real-time execution, finite state machine design, UART, SPI, I2C, sensor fusion, IMU integration, flash data logging, wireless telemetry

Hardware and Electronics: breadboard electronics, PCB layout concepts, circuit wiring and schematic reading, voltage regulation, signal conditioning, H-bridge and servo motor control, ultrasonic and IR sensors, power budgeting, component selection

Mechanical and CAD: Autodesk Fusion 360, structural design, tolerancing, rapid prototyping, 3D printing

Aerospace and Defence: avionics, flight computer design, guidance navigation and control, autonomous recovery systems, defence electronics, high-speed flight systems context (coursework: aerodynamics, thermodynamics, computational modelling)

Methods: hardware-software co-design, systems integration and acceptance testing, fault-finding and debugging, mechatronics, safety-critical firmware development (rocketry flight context)

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Hiring Team Hypersonica

Dear Hiring Team,

I am writing to apply for a summer internship at Hypersonica. Your work on hypersonic propulsion architectures and thermal management for extreme-velocity flight systems represents some of the most technically demanding engineering being pursued today. As a second-year MEng Robotics and Artificial Intelligence student at University College London, I am drawn specifically to Hypersonica's combination of rapid hardware prototyping and advanced computational modelling, and I am eager to contribute to that environment during the summer.

My most directly relevant experience is the development of aerospace avionics firmware for a student rocketry telemetry payload. I designed and implemented C++ firmware on a dual-MCU embedded architecture featuring interrupt-driven real-time execution, I2C sensor integration for IMU data acquisition, altitude sensing, and acceleration logging, and a finite state machine governing autonomous parachute deployment under safety-critical flight conditions. I also implemented wireless telemetry transmission alongside onboard flash data logging. The result was a fully functional flight computer capable of autonomous recovery decisions in a live launch environment, demonstrating my ability to deliver reliable embedded software where failure carries real consequences. This project gave me practical grounding in avionics, guidance and navigation, and hardware-software co-design that I believe translates directly to Hypersonica's engineering challenges.

During this placement I would contribute embedded systems knowledge, a rigorous approach to safety-critical design, and the ability to work across hardware and software simultaneously. I am keen to deepen my understanding of propulsion systems, high-speed aerodynamics, and the engineering trade-offs specific to hypersonic flight regimes. I am confident that working alongside Hypersonica's engineers would accelerate my development considerably, and I would welcome the opportunity to discuss my application further.

Yours sincerely, Aryan Naik